Real-Time Skin Cancer Detection Using YOLOv8 and CNN with Webcam Feed

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***Abstract*—This system performs real-time skin-health monitoring by analyzing live webcam video to detect and classify skin lesions. This model observes skin surface layer-by-layer and identifies irregularity like moles, spots or any pigmented lesions. The system first implements a YOLO-based detector to identify the suspicious lesion area and then applies a Convolutional Neural Network(CNN) classifier to distinguish whether the lesion is normal, suspicious or cancerous(melanoma). The entire system runs using a standard webcam, it makes advanced skin-health accessible to all the users by reducing the dependency on the equipments. Privacy and data protection are the key elements of this system. The combination of YOLO for lesion identification and CNN for classification will make the system more precise than earlier approaches. Its design provides a reliable, real-time, and confidential screening technique that can identify high-risk skin abnormalities early on.**

***Index Terms*—Skin cancer detection, AI analysis, melanoma, YOLOv8**

I. Introduction

One of the fast-growing cancers in the world is skin cancer, and early detection is crucial for reducing death rates and improving the treatment outcome. Manual visual inspection by dermatologists is an indispensable part of traditional diagnosis, but often subjective, and sometimes not available for remote areas. Artificial intelligence innovations have turned automated skin-lesion analysis into a useful tool that raises the bar for early screening accuracy by supporting clinical decisions.The proposed system comprise real-time AI-based image analysis using a standard webcam to constantly monitor the user's lesion on skin and spot potential abnormalities. The architecture consists of two components: a YOLO (You Only Look Once) model responsible for lesion detection and a Convolutional Neural Network (CNN) responsible for classifying the detected region. YOLO will process every frame of the video to find moles or pigmented lesions that look suspicious, perceiving the bounding boxes around the regions of interest. Once a lesion has been isolated, the CNN will classify the extracted region as normal, suspicious, or cancerous. This two-stage pipeline improves accuracy by ensuring that both localization is precise and classification is robust.This system also enables continuous monitoring beyond lesion detection therefore, it has the capability to monitor changes in size, shape, and color over time. Thus, this will make early detection of characteristics related to melanomas, such as asymmetry, border irregularities, and rapid evolution. Its real-time nature provides immediate feedback to the users, instantly helping them identify risks without them having to wait for clinical appointments.Any health-related system has to be concerned with security and integrity. Although the system processes most of the data in real time and does not store any identifiable images, it can incorporate secure communication protocols to protect data when necessary. This ensures the protection of lesion images, results of predictions, and system logs from unauthorized access or manipulation. The system design will be able to support future extension for encrypted storage of lesion history, remote clinical consultation, and automated alerts in case a high-risk abnormality is detected.Deep learning integrated into early detection of skin cancer significantly enhances the performance of manual screening. Thus, with detection by YOLOv8 and classification by CNN, this ensures a reliable, fast, and efficient mechanism for monitoring skin health through a simple webcam by empowering individuals to take proactive steps in recognizing early-stage abnormalities of the skin and seeking immediate medical consultation.

II. Problem statement

Skin cancer remains one of the most common and life-threatening malignancies worldwide, and its prognosis depends on early detection of malignant lesions. Despite the proven benefits of early screening, many people do not have timely access to dermatologists, dermatoscopes, or specialized imaging systems, particularly in rural or resource-limited settings. As a result, after significant enrollment, many cases are diagnosed for which treatment is complex, expensive, and less effective. Advances in deep learning have made it possible to automatically analyze skin lesions with high accuracy in research settings, but these systems typically rely on dermatoscopic images taken under ideal conditions. However, real-world users rely on basic cameras such as laptop or mobile phone webcams, which often suffer from poor lighting, out-of-focus, broken focus, and changes in camera angle or distance. Models trained only on controlled databases often perform poorly in unconstrained environments, limiting their clinical value in the real world. In addition to environmental concerns, existing automated systems often perform semi-tasks (detecting damaged or cropped images or classifying cropped or cropped images) in real time, rather than gradually integrating both steps into the workflow. If the tool is a real-time performance tool, if the tool is designed for live screening, the system must continuously analyze video frames, provide immediate feedback, and operate efficiently on consumer equipment. Object detection architectures such as the YOLO family have shown promising results for corrupting whole images, but most studies focus only on still images and the overall reliability of diagnosing detection latency, frame rate, and video parameters.

This study aims to address this gap in simple, specific sites of skin cancer. The study tests whether these hybrid systems can be delayed, computationally efficient, and maintain clinical accuracy while adapting to uncontrolled environments. Ultimately, what this work aims to achieve is to develop affordable, accessible, real-time tools that can help clinicians and lay users improve early skin cancer screening, early detection capabilities, and patient outcomes.

III. Literature survey

Skin cancer is one of the most common neoplasms diagnosed worldwide, and timely recognition of malignant lesions is important to improve therapeutic efficacy. Traditionally, dermatologists have relied on visual assessments aided by dermoscopy to scrutinize skin lesions for cancerous indications. However, this diagnostic approach can be problematic due to the frequent convergence of visual attributes among various lesions. Recently, the incorporation of artificial intelligence and deep learning methodologies has gained traction in the field of medical image analysis. In particular, convolutional neural networks (CNNs) have exhibited remarkable capabilities in categorizing images relevant to dermatological diseases. A notable study conducted by Esteva demonstrated that deep neural networks can classify skin cancer images with an accuracy comparable to that of experienced dermatologists, underscoring the promise of AI-enabled diagnostic tools [1]. The improvement in automatic recognition of skin lesions has increased significantly due to the availability of large and carefully labeled medical image datasets. A major contribution to this area is the HAM10000 dataset, which was created by Philipp Tschandl. It includes thousands of dermatoscopic images showing different types of pigmented skin lesions. These images were obtained from several different sources. [2] These large datasets allow scientists to train and test deep learning models in controlled experiments. Many studies comparing different methods have shown that deep learning systems work well for diagnosing skin conditions. For example, an investigation by Titus et al.. Brinker et al. reported that deep learning algorithms were better than some dermatologists in identifying melanoma from dermoscopic images [3]. Additional research has indicated that convolutional neural networks can achieve diagnostic precision similar to that of experienced dermatologists in recognizing melanoma and various skin related problems [12]. Researchers have investigated other methods, such as ensemble learning, to improve the effectiveness and dependability of skin lesion assessment [4]. Segmentation and feature extraction methods are vital for accurately pinpointing the edges and structural features of skin lesions in medical images [13][14]. In addition to classification models, object detection algorithms have gained importance in recognizing and pinpointing objects in images. Among these methods, the YOLO framework (You Only Look Once) developed by Joseph Redmon presented a quick and effective technique for the detection of live objects in computer vision applications [6] Over time, several improvements have been introduced to the YOLO architecture to improve its detection performance and computational efficiency. Versions such as YOLOv4 and YOLOv5 incorporate optimized network structures and improved training strategies for real-time detection tasks [7][8]. Recently, YOLOv8, developed by Ultralytics, has provided further improvements in detection accuracy and ease of deployment for modern computer vision applications [9]. Although many existing studies focus on the classification of skin lesions using static images, relatively few systems address real-time detection scenarios. Therefore, combining YOLOv8 for lesion detection with CNN-based classification can enable an effective system for real-time skin cancer detection using webcam feeds, potentially supporting early screening and assisting healthcare professionals in diagnosis of skin cancer.

IV. METHODOLOGY

A. System Architecture

The skin cancer detection system developed in this project features a modular and scalable architecture, combining deep learning–based image analysis with a user-friendly web interface. Each of these four major components constitutes the system: a frontend web application for image acquisition, a backend deep learning inference engine, an image preprocessing module, and a prediction visualization unit. The interaction with the system is provided through the web interface realized using the Streamlit framework, which allows image upload and

real-time webcam capture. The skin lesion image captured or uploaded is sent to the backend module, preprocessed, and passed through a trained CNN classifier. The prediction result would indicate whether the lesion is cancerous or not, and it would be displayed to the user in real time. The modular architecture ensures each module communicates smoothly and easily extends the system for more future enhancements, such as generating medical reports or integrating with a clinical database.

B. Dataset and Preprocessing

The dataset utilized in this system is obtained from the ISIC dataset that comprises dermoscopic images of different skin lesion categories. There are nine different classes involved in the dataset : Actinic Keratosis, Basal Cell Carcinoma, Dermatofibroma, Melanoma , Nevus, Pigmented Benign Keratosis, Seborrheic Keratosis, Squamous Cell Carcinoma, and Vascular Lesions. Extensive pre-processing is done to ensure quality and consistency of data before training the model. All the images are uniformly resized to the fixed resolution 224×224 pixels to make the input dimensions uniform. The pixel values are then normalized between 0 and 1 to increase the convergence while training. Different data augmentations like rotation, horizontal flipping, zooming, and adjusting brightness were performed to increase the diversity of the dataset and to reduce overfitting. The dataset is further divided into a training set and a testing set in a 70:30 ratio to effectively evaluate the performance of the model.

C. AI Model Training and Testing

To classify skin lesions more accurately, the employment of a Convolutional Neural Network is carried out because of its great efficiency in extracting features (both spatial and texture-based) from medical images. The architecture of the CNN model entails various convolutional layers followed by max-pooling layers for capturing hierarchical features, a flattening layer, and finally fully connected dense layers for the classification purpose.The categorical cross-entropy loss and Adam optimizer are used to train the model. During training, performance metrics such as accuracy and loss are monitored continuously in both training and validation datasets. The generalization capability of the trained model is evaluated on unseen test images. After training, the final model is saved in .h5 format and further integrated into the web application for real-time inference. Cancerous classes are grouped to provide the final decision as either “Cancer Detected” or “Not Cancer”.

D. Web-based Image Input and Prediction

Further, a web-based interface is developed that allows the user to interact easily with the system. For this interface, there are two options for input: image upload from local storage and real-time image capture using a webcam. After image submission, the backend CNN model processes the submitted image and instantly shows the prediction result. The output will provide the lesion class, confidence score, and clearly indicate whether the lesion is either cancerous or non-cancerous. This approach, being web-based, facilitates remote screening and preliminary assessment without the need for specialized dermatological equipment.

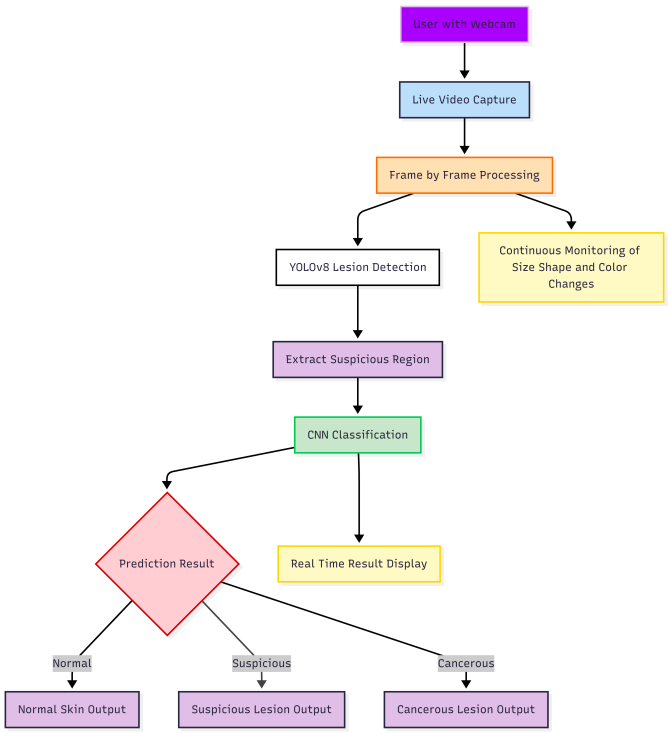

Fig. 1. Skin Cancer Detection

E. Image Processing and Feature Extraction

Preprocessing of images plays an important role in increasing the accuracy of prediction. After acquisition, the input image is converted into RGB format and then resized according to the model’s input shape. The CNN extracts automatically low-level features at early layers, which include edges, color variations, and texture patterns, while deeper layers learn high-level semantic features that are malignant and benign lesion-specific.This automated feature extraction removes the burden of manual feature engineering and enhances the ability of the system to identify subtle visual differences between cancerous and non-cancerous skin lesions.

Fig 1 A webcam records a live video for analysis.YOLOv8 is capable of skin lesion identification and the identification of suspiciousCNN distinguishes the type of lesion. The results are displayed in real time. Output may be normal, suspicious, and/or cancerous.

V. PRELIMINARY RESULTS

The proposed skin cancer detection system uses deep learning methods in the classification of skin lesion images into either cancerous or non-cancerous lesions. A CNN was trained by using a publicly available dataset for skin cancer images

after applying necessary pre-processing steps such as image resizing, normalization, and labeling of classes. The model was saved as .h5 format after training and integrated into a Python-based web application. The system can handle both file input and real-time webcam inputs to analyze the skin lesions with flexibility and user convenience.

Experiments on sample test images showed that the model was able to extract meaningful visual features such as color variation, texture irregularities, and lesion boundaries-an important indicator of skin cancer. The application, created using TensorFlow, Keras, OpenCV, and Streamlit, offers real-time prediction results that require minimum user interaction. Preliminary results obtained from experiments show that the system could form a very good early screening tool in detecting skin cancers. These results are further improvable by training on larger datasets, including multi-class classification, and integrating various state-of-the-art deep learning architectures to enhance the accuracy and reliability of the system.

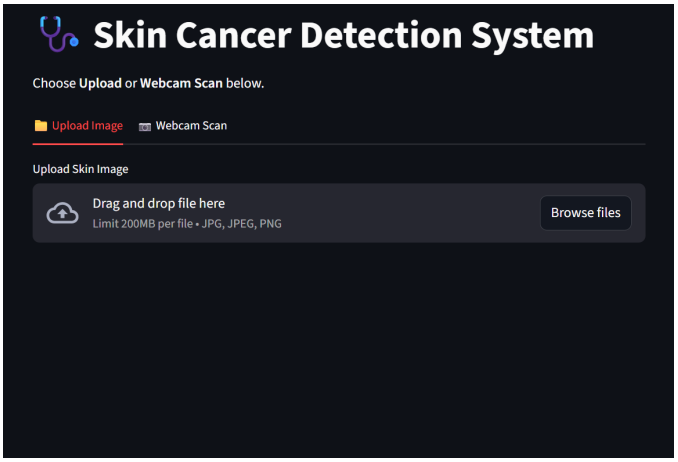

Fig. 2. Skin Cancer Detection System Dashboard

Fig 2 This is an image that shows an online Skin Cancer Detection System that aims at assisting an individual in analyzing skin lesions conveniently. The interface offers the ability to upload an image of the skin condition from the device or take an image from the webcam scanner. This system supports image formats such as JPG, JPEG, and PNG.

Fig 3 This is the interface of a skin cancer detection system after the user has uploaded an image of the skin lesion.It allows two options: image upload and webcam scanning. The interface indicates the selected file,along with a preview of the skin image uploaded.The interface is designed so that a user can easily submit skin images for further analysis regarding whether the lesion is normal or suspicious.

Fig 4 This image signifies the output interface of skin cancer detection system after uploading the image of the skin lesion. This interface indicates that the image has been successfully uploaded, in addition to an alert signal that cancer has been detected. In this system, there is a detailed response to the

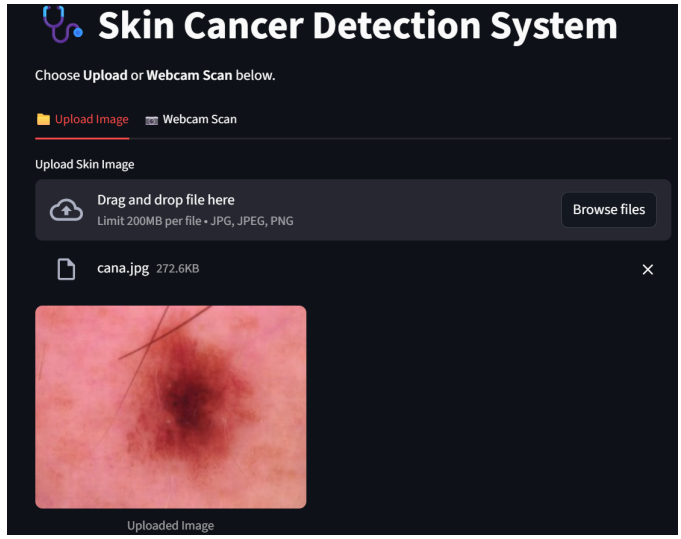

Fig. 3. Lesion Analysis System

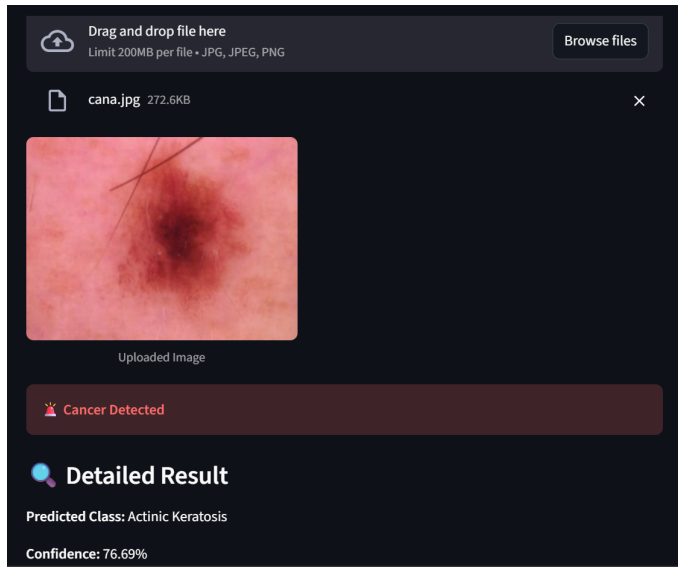

Fig. 4. Cancer Detection Result

classification.This interface signifies the manner in which the skin cancer prediction outcome has been conveyed to the user.

Fig 5 The above image depicts the scanning feature of the webcam in the skin cancer detection system. The live image captured using the webcam is displayed in the captured image section to perform the analysis. The result shows the “Not Cancer” outcome with the confirmation message, with the skin cancer system producing a result that depicts no skin lesion is found. The result section clearly states the result with the predicted class “not applicable” with the confidence level.

Conclusion

The proposed skin cancer detection system provides an efficient method for identifying abnormal skin lesions using

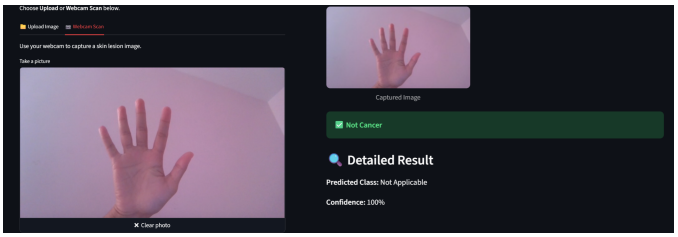

Fig. 5. Real-Time Skin Analysis

image-based analysis. Supporting both image upload and real-time webcam scanning, the system allows for early observation of suspicious skin conditions, which is critical for timely medical intervention. The web interface is designed to be user-friendly, providing ease of access and practical use for preliminary screening.

This will help reduce the effort put into manual examination and improve the results of early diagnosis. With further verification and testing on larger clinical data, the model can be developed further to become a trustworthy supportive tool in dermatological assessment and remote health screening.

FUTUTRE ENHANCEMENTS

Even though advancements are constantly being made, some paths seem to hold great promise for improving the skin cancer detection system's capabilities. Future developments can improve accuracy, usability, and clinical relevance for the benefit of patients and medical professionals, even though the current model can successfully classify skin lesions and support real-time image analysis.

1) Dermatology and Medical Record Systems Integration: By integrating it with hospital databases and electronic medical record systems, the system can be further enhanced by giving dermatologists access to patient lesion histories, historical predictions, and information about lesion developments all in one location. All healthcare data standards, including HL7 and FHIR, would be supported, guaranteeing complete interoperability and enabling multiple clinicians to collaborate remotely and make well-informed diagnostic decisions even faster.

2) Advanced Lesion Analysis and Risk Assessment: In order to evaluate lesion evolution over time, future iterations of the system may incorporate sophisticated image analytical techniques. It may outline differences in a lesion's size, color, or shape and estimate the risk level of a lesion by comparing images from the past and present. These improvements would undoubtedly aid in early detection and prioritize high-risk cases for prompt medical attention.

3) Automated Clinical Referral and Alert Mechanism: In the event of a high-risk detection, this system can be further expanded to automatically suggest dermatological consultation. It might be integrated with telemedicine platforms, allowing for secure lesion image sharing with specialists or instant appointment scheduling. Particularly for patients who reside in remote or underserved areas, this feature will ensure prompt medical interventions and decrease diagnostic delays.

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